Solutions: Optimization

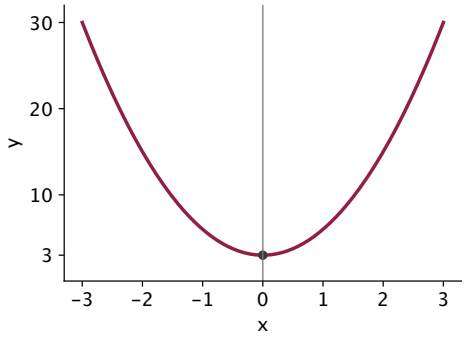
ECON 441: Introduction to Mathematical Economics — Instructor: Div Bhagia

Exercise 9.2

1. (c) $y = 3x^2 + 3$

$$\frac{dy}{dx} = 6x = 0 \implies x^* = 0$$

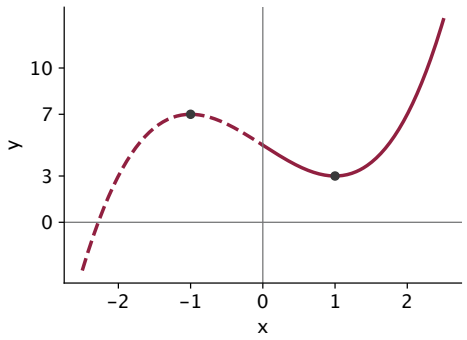
In the immediate neighborhood of 0, $dy/dx < 0$ for $x < 0$ and $dy/dx > 0$ for $x > 0$. The slope changes sign from negative to positive, so the function is decreasing to the left of 0 and increasing to the right: $f(0) = 3$ is a relative minimum. The second derivative confirms this: $d^2y/dx^2 = 6 > 0$.



2. (a) $y = x^3 - 3x + 5$

$$\frac{dy}{dx} = 3x^2 - 3 = 0 \implies x^* = \pm 1$$

The derivative $3(x^2 - 1)$ is negative on the immediate left of 1 (for example at 0.9) and positive on the immediate right (at 1.1), so there is a relative minimum at $x = 1$. It is positive on the immediate left of -1 and negative on the right, so there would be a relative maximum at $x = -1$. But the domain is $[0, \infty)$, so -1 is not admissible. The only stationary value is the relative minimum $f(1) = 3$. The second-derivative test gives the same answer: $d^2y/dx^2 = 6x$, which is $6 > 0$ at $x = 1$.

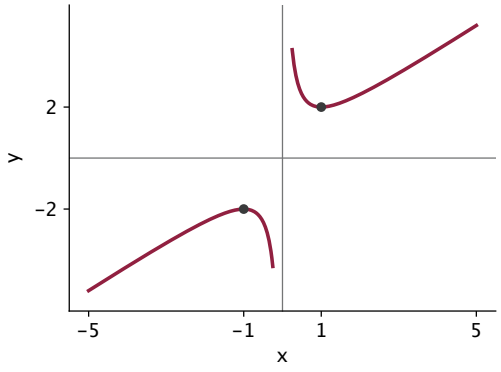


3. $f(x) = x + 1/x$

$$f'(x) = 1 - \frac{1}{x^2} = 0 \;\Rightarrow\; x^* = \pm 1$$

The derivative, $(x^2 - 1)/x^2$, is negative on the immediate left of 1 and positive on the immediate right, so $f(1) = 2$ is a relative minimum. It is positive on the immediate left of -1 and negative on the right, so $f(-1) = -2$ is a relative maximum.

The relative maximum, -2, is lower than the relative minimum, 2. This is not a contradiction: these are *relative* extrema, each compared only with nearby points on its own branch of the graph. The function is not defined at $x = 0$, and the two branches never meet.



4. $T = \phi(x)$

(a) $M = \phi'(x)$ and $A = \phi(x)/x$.

(b) At an extremum of A,

$$A' = \frac{\phi'(x)\,x - \phi(x)}{x^2} = 0 \;\Rightarrow\; \phi'(x^*) = \frac{\phi(x^*)}{x^*}$$

that is, $M = A$.

(c) The marginal curve must cross the average curve at the point where the average curve reaches its extremum (its lowest point for a U-shaped average cost curve).

(d) The elasticity of T is

$$\varepsilon = \frac{\phi'(x)\,x}{\phi(x)} = \frac{M}{A}$$

When $M = A$, the elasticity equals 1.

Exercise 9.4

1. (b) $y = x^3 + 6x^2 + 9$

$$y' = 3x^2 + 12x = 3x(x+4) = 0 \;\Rightarrow\; x^* = 0, -4; \qquad y'' = 6x + 12$$

$y''(0) = 12 > 0$, so $y(0) = 9$ is a relative minimum. $y''(-4) = -12 < 0$, so $y(-4) = 41$ is a relative maximum.

(d) $y = \dfrac{2x}{1-2x}$. By the quotient rule,

$$y' = \frac{2(1-2x) - 2x(-2)}{(1-2x)^2} = \frac{2}{(1-2x)^2}$$

which is never zero. There is no stationary point, so there are no relative extrema. The function is increasing on each side of $x = 1/2$.

2. Let L be the side parallel to the wall and W the two sides perpendicular to it. The netting constraint is $L + 2W = 64$, so $L = 64 - 2W$ and the area is

$$A = LW = (64 - 2W)W = 64W - 2W^2$$

$$\frac{dA}{dW} = 64 - 4W = 0 \;\Rightarrow\; W^* = 16, \quad L^* = 32$$

Since $d^2A/dW^2 = -4 < 0$, this is a maximum. The largest area is $32 \times 16 = 512$ square feet.

3. (a) Yes: $a = 1/3 > 0$, $b = -7 < 0$, $c = 111 > 0$, $d = 50 > 0$, and $b^2 = 49 < 3ac = 111$.

(b) From the demand function, $P = 100 - Q$, so $R = PQ = (100-Q)Q = 100Q - Q^2$.

(c) $\pi = R - C = 100Q - Q^2 - \frac{1}{3}Q^3 + 7Q^2 - 111Q - 50 = -\frac{1}{3}Q^3 + 6Q^2 - 11Q - 50$

(d)
$$\frac{d\pi}{dQ} = -Q^2 + 12Q - 11 = 0 \;\Rightarrow\; (Q-1)(Q-11) = 0 \;\Rightarrow\; Q = 1 \text{ or } 11$$
$\frac{d^2\pi}{dQ^2} = -2Q + 12$ is $10 > 0$ at $Q = 1$ (a minimum) and $-10 < 0$ at $Q = 11$ (a maximum). So $Q^* = 11$.

(e) $\pi(11) = -\frac{1}{3}(1331) + 6(121) - 11(11) - 50 = 334/3 \approx 111.33$.

Exercise 9.3

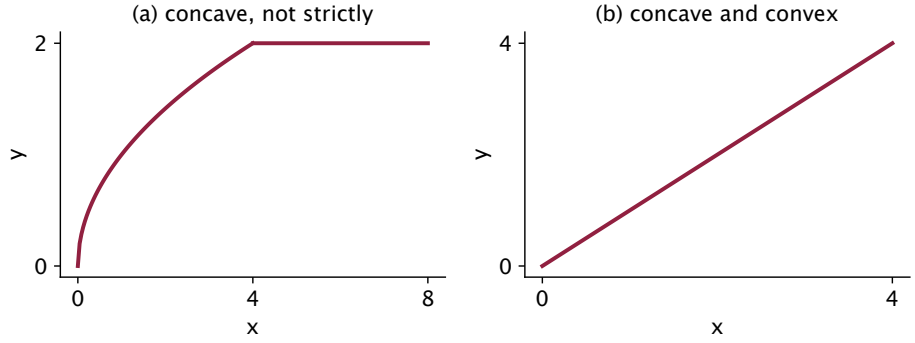
2. (a) $y = 9x^2 - 4x + 8$: $y' = 18x - 4$ and $y'' = 18 > 0$ for all x, so strictly convex.

   (b) $w = -3x^2 + 39$: $w' = -6x$ and $w'' = -6 < 0$, so strictly concave, not convex.

   (c) $u = 9 - 2x^2$: $u' = -4x$ and $u'' = -4 < 0$, so strictly concave, not convex.

   (d) $v = 8 - 5x + x^2$: $v' = -5 + 2x$ and $v'' = 2 > 0$, so strictly convex.

3. (a) Any concave curve that contains a straight segment, for example $y = \sqrt{x}$ for $0 \le x \le 4$ continued as the horizontal line $y = 2$ for $x > 4$. (b) A straight line, for example $y = x$: its second derivative is zero everywhere, so it satisfies both $f'' \le 0$ and $f'' \ge 0$.



4. $y = a - \dfrac{b}{c + x}$

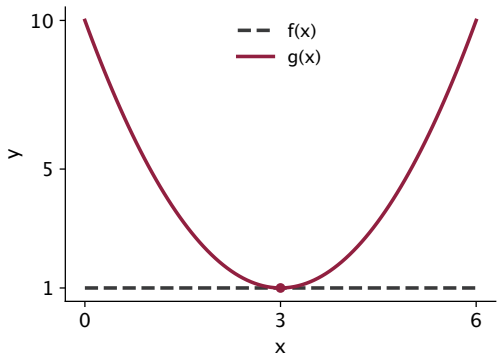
   (a) $\dfrac{dy}{dx} = \dfrac{b}{(c+x)^2} > 0$ and $\dfrac{d^2y}{dx^2} = -\dfrac{2b}{(c+x)^3} < 0$, so the function is increasing and strictly concave.

   (b) At $x = 0$, $y = a - b/c$.

   (c) As $x \to \infty$, $b/(c+x) \to 0$, so $y \to a$. The graph rises from $a - b/c$ toward the horizontal asymptote $y = a$, flattening as it goes.

As a consumption function with income x: consumption should be positive at zero income, which requires $ac > b$, and the marginal propensity to consume dy/dx should be less than 1 at every income level, which requires $b/c^2 < 1$, that is, $b < c^2$.

5. $f(x)$ must be a constant, say $f(x) = 1$; every point is a stationary point. $g(x)$ could be $g(x) = x^2 - 6x + 10$, which has a single stationary point at $x = 3$. The difference: f is stationary everywhere, whereas g is stationary only at $x = 3$.



Exercise 9.4

5. $\pi(Q) = hQ^2 + jQ + k$

   (a) $\pi(0) = k$, so $k < 0$.

   (b) $\pi'(Q) = 2hQ + j$ and $\pi''(Q) = 2h$, so strict concavity requires $h < 0$.

   (c) The critical point is $Q^* = -j/(2h)$. With $h < 0$, we need $j > 0$ to make $Q^* > 0$.

Exercise 11.2

1. $z = x^2 + xy + 2y^2 + 3$

   The first-order conditions are

$$z_x = 2x + y = 0 \quad \text{and} \quad z_y = x + 4y = 0$$

   which give $(x^*, y^*) = (0, 0)$, with $z^* = 3$. At this point,

$$z_{xx} = 2 > 0, \quad z_{yy} = 4 > 0, \quad z_{xy} = 1, \quad z_{xx}z_{yy} = 8 > 1 = z_{xy}^2$$

   so z^* is a minimum.

2. $z = -x^2 - y^2 + 6x + 2y$

   The first-order conditions are

   $$z_x = -2x + 6 = 0 \quad \text{and} \quad z_y = -2y + 2 = 0$$

   which give $(x^*, y^*) = (3, 1)$, with $z^* = 10$. At this point,

   $$z_{xx} = -2 < 0, \quad z_{yy} = -2 < 0, \quad z_{xy} = 0, \quad z_{xx}z_{yy} = 4 > 0 = z_{xy}^2$$

   so z^* is a maximum.

3. $z = ax^2 + by^2 + c$

   The first-order conditions $z_x = 2ax = 0$ and $z_y = 2by = 0$ give $(x^*, y^*) = (0, 0)$, with $z^* = c$. Here $z_{xx} = 2a$, $z_{yy} = 2b$, $z_{xy} = 0$, and $z_{xx}z_{yy} = 4ab$.

   (a) With $a, b > 0$, $z_{xx} > 0$, $z_{yy} > 0$, and $4ab > 0$, so $z^* = c$ is a minimum.

   (b) With $a, b < 0$, $z_{xx} < 0$, $z_{yy} < 0$, and $4ab > 0$, so $z^* = c$ is a maximum.

   (c) With a and b opposite in sign, $4ab < 0 = z_{xy}^2$, so $(0, 0)$ is a saddle point: neither a maximum nor a minimum.

4. $z = e^{2x} - 2x + 2y^2 + 3$

   The first-order conditions are

   $$z_x = 2e^{2x} - 2 = 0 \quad \text{and} \quad z_y = 4y = 0$$

   which give $(x^*, y^*) = (0, 0)$, with $z^* = 4$. At this point,

   $$z_{xx} = 4e^{2x} = 4 > 0, \quad z_{yy} = 4 > 0, \quad z_{xy} = 0, \quad z_{xx}z_{yy} = 16 > 0 = z_{xy}^2$$

   so z^* is a minimum.

5. $z = (x - 2)^4 + (y - 3)^4$

   (a) $(x-2)^4$ and $(y-3)^4$ are never negative, and both are zero only at $x = 2$, $y = 3$. So $z \geq 0$ everywhere, with $z^* = 0$ at $(2, 3)$.

   (b) Yes. $z_x = 4(x - 2)^3$ and $z_y = 4(y - 3)^3$ are both zero at $(2, 3)$.

   (c) No. $z_{xx} = 12(x - 2)^2$, $z_{yy} = 12(y - 3)^2$, and $z_{xy} = 0$ are all zero at $(2, 3)$, so $z_{xx} > 0$ fails. The second-order condition is sufficient, not necessary: this point is a minimum even though the condition does not hold.

Exercise 12.2

1. Use the Lagrange-multiplier method to find the stationary values of z.

   (a) $z = xy$, subject to $x + 2y = 2$

   The Lagrangian is

   $$L(x, y, \lambda) = xy + \lambda(2 - x - 2y)$$

   with first-order conditions

   $$L_x = y - \lambda = 0, \qquad L_y = x - 2\lambda = 0, \qquad L_\lambda = 2 - x - 2y = 0$$

   The first two give $y = \lambda$ and $x = 2\lambda$, so $x = 2y$. Substituting into the constraint, $2 - 2y - 2y = 0$, so $y^* = \frac{1}{2}$ and $x^* = 1$. Then $\lambda^* = y^* = \frac{1}{2}$ and $z^* = \frac{1}{2}$.

   (b) $z = x(y + 4)$, subject to $x + y = 8$

   The Lagrangian is

   $$L(x, y, \lambda) = x(y + 4) + \lambda(8 - x - y)$$

   with first-order conditions

   $$L_x = y + 4 - \lambda = 0, \qquad L_y = x - \lambda = 0, \qquad L_\lambda = 8 - x - y = 0$$

   The first two give $x = y + 4$. Substituting into the constraint, $8 - (y + 4) - y = 0$, so $y^* = 2$ and $x^* = 6$. Then $\lambda^* = x^* = 6$ and $z^* = 36$.

   (c) $z = x - 3y - xy$, subject to $x + y = 6$

   The Lagrangian is

   $$L(x, y, \lambda) = x - 3y - xy + \lambda(6 - x - y)$$

   with first-order conditions

   $$L_x = 1 - y - \lambda = 0, \qquad L_y = -3 - x - \lambda = 0, \qquad L_\lambda = 6 - x - y = 0$$

   The first two give $1 - y = -3 - x$, so $y = x + 4$. Substituting into the constraint, $6 - x - (x + 4) = 2 - 2x = 0$, so $x^* = 1$ and $y^* = 5$. Then $\lambda^* = 1 - y^* = -4$ and $z^* = -19$.

   (d) $z = 7 - y + x^2$, subject to $x + y = 0$

The Lagrangian is

$$L(x, y, \lambda) = 7 - y + x^2 + \lambda(0 - x - y)$$

with first-order conditions

$$L_x = 2x - \lambda = 0, \qquad L_y = -1 - \lambda = 0, \qquad L_\lambda = -x - y = 0$$

The second gives $\lambda^* = -1$; the first then gives $x^* = \lambda^*/2 = -\frac{1}{2}$, and the constraint gives $y^* = -x^* = \frac{1}{2}$. So $z^* = 7 - \frac{1}{2} + \frac{1}{4} = \frac{27}{4}$.

2. In Prob. 1, find whether a slight relaxation of the constraint will increase or decrease the optimal value of z. At what rate?

   Write the constraint as $g(x, y) = c$. At the optimum, $dz^*/dc = \lambda^*$: by the chain rule,

$$\frac{dz^*}{dc} = f_x \frac{dx^*}{dc} + f_y \frac{dy^*}{dc} = \lambda^* \left(g_x \frac{dx^*}{dc} + g_y \frac{dy^*}{dc} \right) = \lambda^*$$

   using $f_x = \lambda^* g_x$ and $f_y = \lambda^* g_y$ from the first-order conditions, and the fact that differentiating the constraint $g(x^*, y^*) = c$ with respect to c makes the term in parentheses equal 1. So relaxing the constraint (raising c slightly) changes z^* at the rate λ^*:

   (a) $\lambda^* = \frac{1}{2}$: z^* increases at rate $\frac{1}{2}$.

   (b) $\lambda^* = 6$: z^* increases at rate 6.

   (c) $\lambda^* = -4$: z^* decreases at rate 4.

   (d) $\lambda^* = -1$: z^* decreases at rate 1.

3. Write the Lagrangian function and the first-order condition for stationary values (without solving the equations) for each of the following:

   (a) $z = x + 2y + 3w + xy - yw$, subject to $x + y + 2w = 10$

$$L = x + 2y + 3w + xy - yw + \lambda(10 - x - y - 2w)$$

First-order conditions:

$$\begin{aligned} L_x &= 1 + y - \lambda = 0 \\ L_y &= 2 + x - w - \lambda = 0 \\ L_w &= 3 - y - 2\lambda = 0 \\ L_\lambda &= 10 - x - y - 2w = 0 \end{aligned}$$

(b) $z = x^2 + 2xy + yw^2$, subject to $2x + y + w^2 = 24$ and $x + w = 8$

With one multiplier per constraint,

$$L = x^2 + 2xy + yw^2 + \lambda(24 - 2x - y - w^2) + \mu(8 - x - w)$$

First-order conditions:

$$\begin{aligned} L_x &= 2x + 2y - 2\lambda - \mu = 0 \\ L_y &= 2x + w^2 - \lambda = 0 \\ L_w &= 2yw - 2w\lambda - \mu = 0 \\ L_\lambda &= 24 - 2x - y - w^2 = 0 \\ L_\mu &= 8 - x - w = 0 \end{aligned}$$

4. If, instead of $g(x, y) = c$, the constraint is written in the form of $G(x, y) = 0$, how should the Lagrangian function and the first-order condition be modified as a consequence?

   The constant is now zero, so the Lagrangian becomes

$$L(x, y, \lambda) = f(x, y) + \lambda[0 - G(x, y)] = f(x, y) - \lambda G(x, y)$$

   and the first-order conditions are

$$L_x = f_x - \lambda G_x = 0, \qquad L_y = f_y - \lambda G_y = 0, \qquad L_\lambda = -G(x, y) = 0$$